



Heat and mass transfer modeling and performance analysis of a multi-stage coaxial solar-magnetic hybrid distiller based on global energy balance

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ABSTRACT

Dissipation of condensation latent heat constrains the energy utilization, and multi-stage latent heat recovery is a promising strategy to overcome the efficiency limitation of solar-driven interfacial evaporation. Here, we design and demonstrate a multi-stage coaxial solar-magnetothermal distiller (MSCSMD) employing a three dimensional magnetic porous carbon scaffold for heat generation, enabling efficient freshwater production under photo-thermal and magnetothermal heating. To elucidate the coupled heat and mass transfer within MSCSMD and predict the evaporation performance, a radial heat and mass transfer model is developed based on global energy balance and solved using a backward iteration scheme. Firstly, experimental validation confirms good agreement between predictions and measurements, with relative deviations of 2–7% in outer wall temperature and water yield. Benefiting from stage wise recovery of condensation latent heat in coaxially nested units, the 16-stage MSCSMD achieves a recoverable efficiency of 581.42% and a nominal evaporation rate of 65.99 kg·m⁻²·h⁻¹. Based on parameter sensitivity analysis, the gap width should be constrained to $\delta_{gap} < 5$ mm to ensure efficient heat transfer and to mitigate the rapid escalation of sidewall losses and vapor diffusion resistance. Moreover, a longer axial length strengthens heat recovery and diminishes sidewall losses, increasing the total efficiency of the 16-stage device to 779.77%, corresponding to a GOR of 7.80, confirming the advantage of high aspect-ratio designs in retaining heat. Increasing ambient temperature or reducing external heat transfer coefficient suppresses convective and radiative losses, shifts operation to a higher temperature condition and enhances water production. Finally, economic analysis reveals a marginal electricity cost of 0.043 RMB·L⁻¹ for freshwater production in the solar-magnetothermal mode. This work provides mechanistic insights into heat and mass transport in coaxial multi-stage distillers, and offers a novel model for the design and performance prediction of compact, high efficiency distillation systems.

1. Introduction

Freshwater scarcity has become one of the most pressing global challenges, and the tension between supply and demand is further intensified by population growth, industrialization, and climate change [1,2]. Conventional desalination technologies, such as reverse osmosis (RO), multi-effect distillation (MED), and multi-stage flash desalination (MSF), have been widely used to expand freshwater supply [3]. Nevertheless, their dependence on continuous energy input, large-scale infrastructure, and complex system operation limits their broader application in decentralized and energy-constrained scenarios [4]. In this regard, interfacial solar steam generation (ISSG) has garnered significant interest as a sustainable and low-carbon solution for seawater

desalination, owing to its ability to localize heat at the water-air interface, thereby minimizing sensible heat loss to the bulk water [5–7]. Substantial progress has been made by developing high performance photothermal materials, including metallic nanoparticles [8–10], semiconductors [11–13], and carbon based absorbers [14–16], together with improved water transport architectures [17,18] and three dimensional evaporator designs [19,20], leading to markedly enhanced evaporation rates. Nevertheless, the freshwater yield of conventional single stage devices remains fundamentally constrained by the large latent heat of vaporization (2450 kJ·kg⁻¹). Under one sun irradiation, the theoretical evaporation limit is only 1.4–1.6 kg·m⁻²·h⁻¹, and the energy utilization efficiency is inherently capped below 100% when referenced to the external input [21,22]. Therefore, recovering the substantial latent heat released during vapor condensation is essential to

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Nomenclature*Latin Letters*

A	Area m^2
D_a	Diffusion coefficient of vapor in air m^2/s
h_a	Convection heat transfer coefficient $W/(m^2 \cdot K)$
h_{fg}	Latent heat of evaporation J/kg
J	Mass flow rate of vapor kg/s
k	Thermal conductivity $W/(m \cdot K)$
L	Length of the cylinder m
M_w	Molar mass of water kg/mol
n	Total number of stages
p_{sat}	Saturation vapor pressure Pa
Q	Heat flow rate/Power W
r	Radius m
R_u	Universal gas constant $J/(mol \cdot K)$
R_{th}	Thermal resistance K/W
T	Temperature K or $^{\circ}C$
v	Evaporation rate $kg/(m^2 \cdot h)$

Greek Letters

δ	Thickness or gap width m
ε	Emissivity
η	Efficiency $\%$
σ	Stefan-Boltzmann constant $W/(m^2 \cdot K^4)$

Subscripts

act	Actual /
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$cond$	Conduction /
$evap$	Evaporation /
ext	External /
gap	Vapor gap/
gas	Gas/
i	i-th stage/
in	Input/
ins	Insulation wall
mag	Magnetic
nom	Nominal /
out	Output /
rad	Radiation
rec	Recovery
sun	Solar
$side$	Side wall
tot	Total
$wall$	Out wall
e	Evaporation-side interface
c	Condensation-side interface

Abbreviations

AMF	Alternating Magnetic Field/
$ISSG$	Interfacial Solar Steam Generation/
$MSCMD$	Multi-Stage Coaxial Solar-Magnetic Distiller
MSD	Multi-Stage Distillation
GOR	Gained Output Ratio/

transcend the single stage ceiling and to enable high performance desalination.

In traditional single-stage ISSG systems, vapor generated at the photothermal interface diffuses across an air gap and condenses on a cold wall for freshwater collection [23]. A fundamental thermodynamic drawback, however, is that the latent heat released upon condensation is usually dissipated as waste heat into the environment, resulting in irreversible energy loss [24,25]. Multi-stage distillation (MSD) structures are designed to improve heat utilization by harvesting and reusing condensation latent heat. In a typical evaporation and condensation sequence, vapor generated in stage (i) condenses on the adjacent condensation surface, and the released latent heat is subsequently transferred to drive evaporation in stage ($i + 1$). This staged heat reuse mechanism has been widely adopted in conventional thermal desalination technologies, such as multi-effect distillation (MED) and multi-stage flash (MSF), and has also inspired recent multi-stage solar distillers [26,27]. However, achieving efficient latent-heat recovery in compact interfacial solar desalination systems remains challenging because the heat input is highly localized, the available temperature difference is limited, and heat losses become increasingly significant at small device scales.

Recent solar-driven multi-stage distillation systems have attempted to miniaturize the latent-heat recovery principle by coupling evaporation and condensation within compact photothermal architectures. Among the evolving MSD configurations, three dimensional stacked distillers have been most widely explored. Yang *et al.* [28] reported a passive multi-stage solar distiller that also enabled directional crystallization separation, achieving a water yield of $1.7 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ with a reported energy efficiency of 114% in a three-stage device. From a theoretical perspective, Zhang *et al.* [29] established a comprehensive framework to model coupled heat and mass transfer in heat localized multi-stage solar distillers, and predicted efficiencies exceeding 700% by geometric optimization. Li *et al.* [30] developed a foldable three dimensional structure that enhanced light harvesting and air

convection, delivering an evaporation rate as high as $3.22 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ in a seven-stage device. Wen *et al.* [31] further demonstrated an asymmetric conical multi-stage distiller based on an optimized balance between evaporation and condensation mass transfer, leading to exceptionally high freshwater productivity. Despite such progress, the planar structure of stacked systems inevitably suffers from a large spatial footprint. Just as Zhang *et al.* [32] have studied and found, severe convective and radiative heat losses at the exposed edges act as a bottleneck, restricting further efficiency gains in compact applications. In addition, the evaporation and condensation capacities in planar stacked devices may not always be well matched, leading to underutilized condensation surfaces or insufficient vapor removal in downstream stages.

To alleviate the space penalty and edge losses associated with planar designs, circumferential or coaxial MSD concepts have emerged as more compact and potentially more efficient alternatives. By radially nesting cylindrical evaporation-condensation units, a coaxial structure can maximize the specific evaporation area within a minimal footprint, while the radial temperature gradient helps suppress heat leakage to the environment. Zhao *et al.* [33], inspired by floral morphology, developed a coaxial shell type solar distiller and established heat and mass transfer models for both single-stage and multi-stage systems. Through structural optimization, they reported a total energy efficiency of 480%, corresponding to a 10.66 times improvement over conventional solar distillation systems (45%). Guided by this concept, coaxial distillers differ from previously reported planar stacked configurations by redirecting the main transport pathway from vertical layer-to-layer transfer to radial vapor generation, diffusion, and condensation. In this geometry, the vapor diffusion area, cylindrical conduction resistance, radiative heat exchange, and external wall heat loss all change with the radial position and stage number. As a result, the heat and mass transfer processes become more strongly coupled and require dedicated modeling. Nevertheless, even with favorable geometry, reliance on intermittent solar irradiation remains a practical barrier to continuous freshwater

production. To compensate for the variability of solar input, additional energy sources have been incorporated. For example, Poredoš *et al.* [34] proposed a stage temperature boosting (STB) concept that routes low grade heat to the final-stage condenser, strengthening vapor generation and condensation across stages and improving performance by 88% relative to a 10-stage solar MSD device. Chen *et al.* [35] implemented a hybrid strategy in which part of the solar energy drives multi-stage photothermal distillation while the remainder is converted and stored via photovoltaics for subsequent electrothermal distillation during sunless periods, enabling stable and continuous freshwater production of $2.5 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$. In addition to these strategies, wave energy [36], wind energy [37], industrial low-grade waste heat [38], and magnetothermal heating [39,40] have also been explored to enhance interfacial evaporation, collectively highlighting the value of external energy input in boosting evaporation intensity and enabling continuous operation. Among these auxiliary strategies, magnetothermal heating provides controllable and localized heat compensation through magnetic porous materials. Under an alternating magnetic field, heat can be generated non-contact within the evaporator, thereby supplementing solar input under weak or fluctuating irradiation and improving the operating flexibility of solar-driven desalination systems [41]. From the perspective of interfacial transport, Xu *et al.* [42] investigated solar-driven saline evaporation from an open meniscus in a capillary tube and highlighted the importance of coupled heat and mass transfer near the evaporation interface in regulating evaporation behavior. These findings emphasize the need to properly resolve interfacial heat and vapor transport in confined evaporation systems, particularly when multiple evaporation–condensation interfaces are coupled within a compact geometry. Despite escalating experimental endeavors, a comprehensive theoretical understanding of heat and mass transfer within compact 3D multi-stage distillers, especially those leveraging external fields, is still lacking. Consequently, the absence of high fidelity models acts as a critical bottleneck, impeding precise design, scalability, and structural optimization.

Motivated by these challenges, this work develops a multi-stage coaxial solar-magnetothermal distiller (MSCSMD) for compact radial latent heat recovery. However, unlike planar stacked distillers, the proposed device arranges evaporation and condensation units in a coaxially nested configuration, which redirects vapor transport into a radial evaporation, diffusion, and condensation process and improves spatial integration. Besides, a three-dimensional magnetic porous carbon scaffold is introduced as an integrated photothermal and magnetothermal heat-generation medium, enabling localized solar absorption and controllable magnetic heat compensation under an alternating magnetic field. In addition, a radial heat and mass transfer model based on global energy balance is established and solved using a backward iterative algorithm, allowing the coupled effects of vapor diffusion, cylindrical heat conduction, radiation exchange, sidewall heat loss, and interstage latent heat transfer to be quantified. The model is validated using experimental data and is further applied to evaluate the effects of gap width, axial length, stage number, ambient temperature, and external heat transfer coefficient on freshwater production and energy utilization. This work provides a modeling framework and design guidance for compact multi-stage distillation systems with coupled solar and magnetothermal heat input.

2. Theoretical analysis

This section presents the concept and working principle of the multi-stage coaxial solar-magnetothermal distiller, together with the design and fabrication of the geometric structure. On this basis, heat and mass transfer models are developed for both single-stage and multi-stage evaporation–condensation processes, and the modeling accuracy is validated through experiments.

2.1. Working principle

A multi-stage coaxial solar-magnetothermal distiller (MSCSMD) is designed in this work, and the heat and mass transfer and energy cycling process are schematically illustrated in Fig. 1. A three-dimensional magnetic porous carbon scaffold placed at the center serves as an integrated photothermal and magnetothermal heat source, converting solar irradiation and alternating magnetic field excitation into localized heat that continuously supplies the evaporation layer. Driven by capillary pumping in an ultra-hydrophilic wick, seawater is steadily and uniformly delivered to the central evaporation layer, where rapid phase change occurs under the localized high temperature. The generated vapor diffuses radially across the vapor gap and condenses on the condensation layer to produce freshwater, releasing latent heat. Instead of being dissipated to the environment, the released heat is predominantly conducted through a low thermal resistance thin wall to the hydrophilic evaporator in the next stage, providing a secondary heat source for continued evaporation. This repeated evaporation and condensation units realizes stage-wise latent heat recovery and sustained freshwater output, resulting in markedly enhanced water yield and energy recovery.

2.2. Distiller fabrication and testing

Fig. 2 depicts the geometric design, fabrication, and the photothermal and magnetothermal test platform of the MSCSMD. As shown in Fig. 2a–b, the distiller adopts a highly integrated coaxial, multi-stage cylindrical design, in which the evaporation–condensation layers are radially nested in a stage-wise manner. A three-dimensional magnetic porous carbon scaffold is positioned at the center and serves as the heat-generation medium, and this coaxial layout is intended to maximize radial space utilization while suppressing heat loss. Guided by the geometric modeling and structural design, the stage components with resin substrates were manufactured via high-precision 3D printing (Fig. 2c). Accurate control of the micro scale gap width, together with the mechanical stability of the printed parts, ensures reliable assembly and structural robustness during operation. To evaluate freshwater production under combined solar irradiation and an alternating magnetic field (AMF), an evaporation–condensation test platform was constructed (Fig. 2d–e). The MSCSMD was placed on a reservoir containing simulated seawater, and continuous water supply was maintained through a hydrophilic wick at the bottom, and the condensed freshwater was collected for quantification. In terms of energy input, a xenon lamp solar simulator provided normal (vertical) irradiation from top direction, while a surrounding high frequency induction copper coil generated the AMF to enable synergistic photothermal and magnetothermal heating. During testing, a high resolution infrared (IR) camera was used to record the surface temperature field and the dynamic thermal response of each stage, providing experimental validation of the temperature distribution.

2.3. Theoretical model development

2.3.1. Single-stage heat and mass transfer model

As shown in Fig. 3a, a steady-state heat and mass transfer model is established for a single-stage coaxial evaporation–condensation unit, with the inner cylinder as the evaporation layer and the outer cylinder as the condensation layer separated by a vapor gap. Combined photothermal and magnetothermal inputs drive evaporation, the generated vapor diffuses radially across the gap and condenses on the cooler wall, releasing latent heat. The model is established based on the following assumptions: (1) the system operates under a steady state; (2) heat and mass transfer in each stage are dominated by the radial direction, while axial gradients within the evaporation–condensation gap are neglected; (3) natural convection in the vapor gap is negligible; (4) water vapor is treated as the only diffusing component, and the influence of salinity on

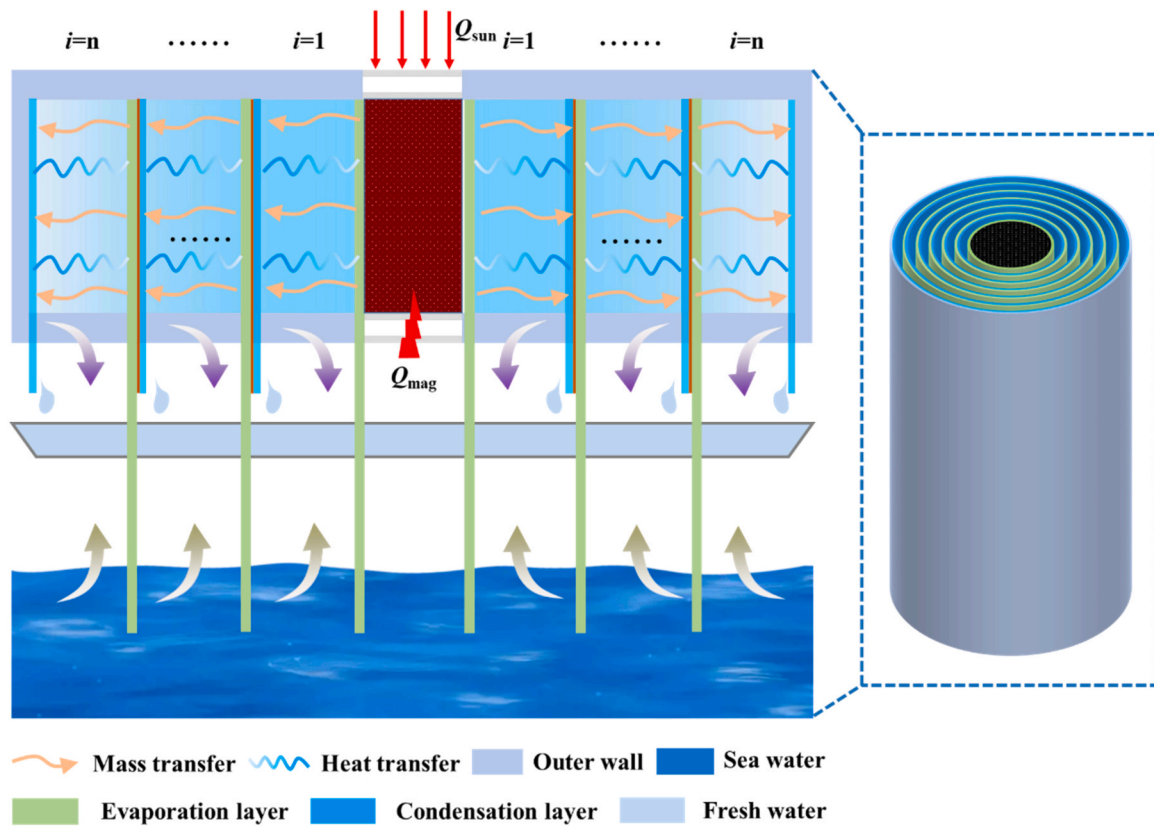


Fig. 1. Schematic illustration of the concept and operating principle of the MSCSMD.

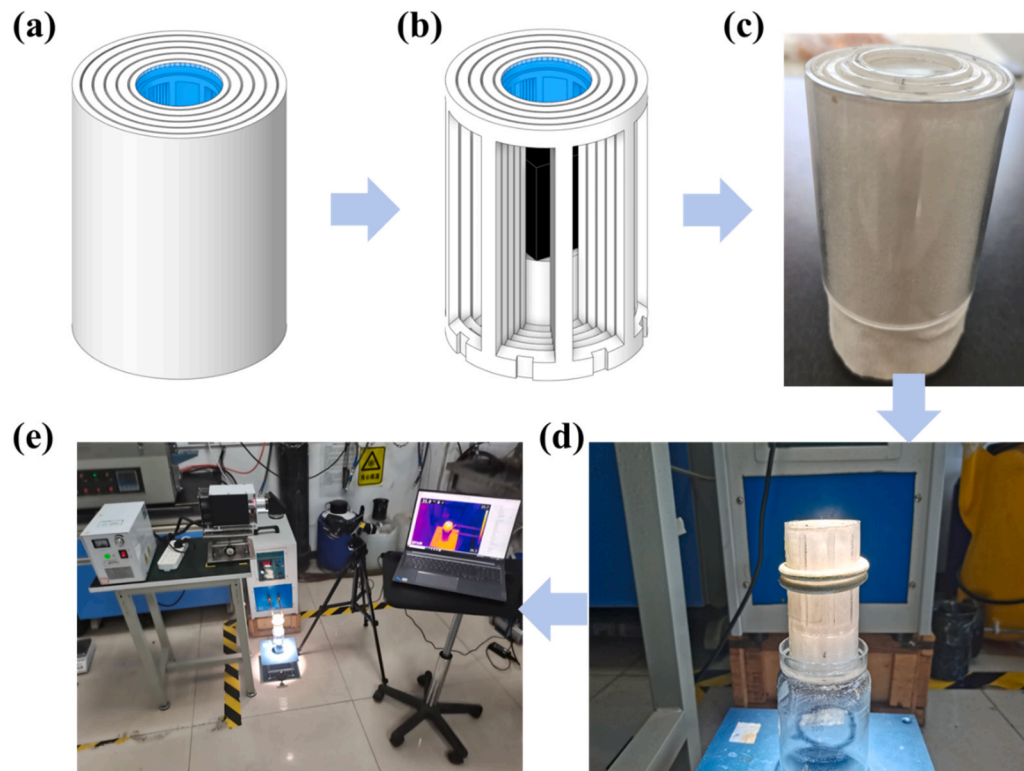


Fig. 2. Geometric structure of the MSCSMD and the experimental test platform. (a-b) Geometric modeling and device design of the MSCSMD. (c) Photograph of the MSCSMD prototype fabricated by 3D printing and assembly. (d) MSCSMD operated under combined solar irradiation and an alternating magnetic field (AMF) generated by the induction coil. (e) Schematic of the evaporation–condensation test platform coupling solar irradiation with AMF heating.

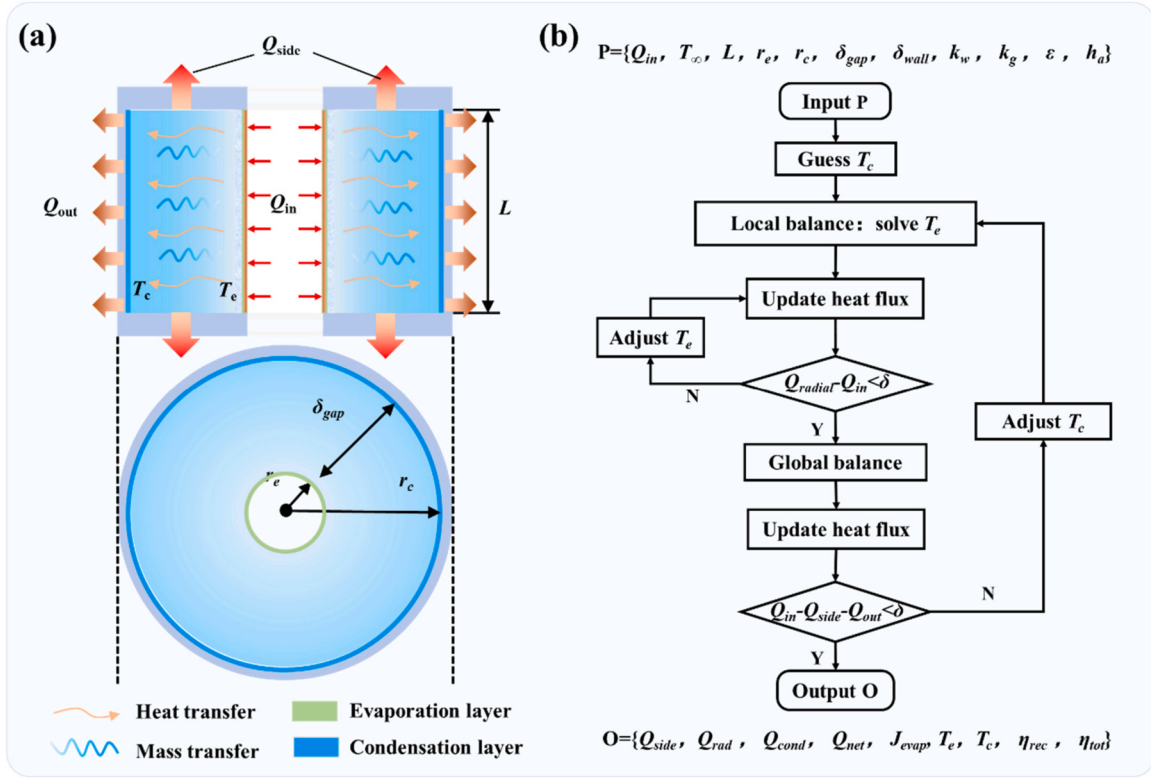


Fig. 3. Single-stage heat and mass transfer model and solution process for the coaxial cylindrical evaporator. (a) Schematic of the heat and mass transfer model. (b) Flowchart of the numerical algorithm used to solve the coupled heat and mass transfer equations.

vapor pressure is neglected; (5) the evaporation and condensation layers are sufficiently thin, and their conductive thermal resistances are negligible compared with that of the vapor gap; and (6) no vapor or liquid leakage occurs. The validity of these assumptions is further assessed in Notes S1–S3 [43]. Global energy conservation and radial energy transport are then formulated, and the energy balance equation is:

$$Q_{in} = Q_{loss,out} + Q_{loss,side} \quad (1)$$

Where Q_{in} is the total input power, consisting of the photothermal contribution Q_{sun} and the magnetothermal contribution Q_{mag} , $Q_{loss,out}$ and $Q_{loss,side}$ denote the heat losses through the outer wall and the side-wall.

Under steady-state conditions, the energy supplied to the inner wall is transferred radially across the vapor gap to the outer wall. The radial heat flux Q_{radial} can be divided into three parts:

$$Q_{in} = Q_{radial} = Q_{evap} + Q_{cond} + Q_{rad} \quad (2)$$

where Q_{evap} denotes the latent-heat term associated with evaporation, and Q_{cond} and Q_{rad} represent the heat transferred across the gap via conduction and radiation.

Based on the Stefan-flow diffusion theory, with a geometric correction for the radius dependent diffusion area in cylindrical coordinates, the vapor mass diffusion flux and the associated latent heat term can be expressed as [44]:

$$J_{evap} = \frac{D_a M_w}{R_u T_{avg}} \cdot \frac{2\pi L}{\ln(r_c/r_e)} \cdot [P_{sat}(T_e) - P_{sat}(T_c)] \quad (3)$$

$$Q_{evap} = J_{evap} \cdot h_{fg} \quad (4)$$

where D_a is the diffusion coefficient of vapor, M_w is the molar mass of water, and R_u is the universal gas constant, r_e and r_c denote the radius of

evaporation interface and condensation interface, and T_e and T_c represent the corresponding temperatures. It should be noted that T_e and T_c are not prescribed boundary temperatures; instead, they are numerically determined from the coupled local radial and global energy-balance equations using the iterative procedure described below. $P_{sat}(T)$ is the saturation vapor pressure, which is evaluated using the Antoine equation [45].

Heat conduction across the vapor gap is described by the one-dimensional Fourier law for coaxial cylindrical walls:

$$Q_{cond} = \frac{2\pi L \cdot k_{gas}(T_e - T_c)}{\ln(r_c/r_e)} \quad (5)$$

where k_{gas} is the thermal conductivity of the gas mixture in the gap.

Radiative heat transfer across the vapor gap can be expressed as:

$$Q_{rad} = \frac{\sigma A (T_e^4 - T_c^4)}{\frac{1}{\epsilon_e} + \frac{1 - \epsilon_c}{\epsilon_c} \left(\frac{r_e}{r_c}\right)} \quad (6)$$

Where σ is the Stefan–Boltzmann constant, ϵ is the surface emissivity, and A is the heat-transfer area.

Third-kind boundary conditions are imposed on the outer wall, and their surface temperatures are determined from node-wise energy balances:

$$\frac{T_c - T_{s,out}}{R_{th,wall}} = h_a A_{out} (T_{s,out} - T_\infty) + \epsilon_{ext} \sigma A_{out} (T_{s,out}^4 - T_\infty^4) \quad (7)$$

The conductive thermal resistance of the outer-wall insulation layer is given by:

$$R_{th,wall} = \frac{\ln((r_c + \delta_{wall})/r_c)}{2\pi L k_{wall}} \quad (8)$$

The heat loss through the outer wall is given by:

$$Q_{loss,out} = \frac{T_c - T_{s,out}}{R_{th,wall}} \quad (9)$$

where $T_{s,out}$ is the outer surface temperature, h_a is the natural convection heat-transfer coefficient, A_{out} is the outer surface area, and δ_{wall} and k_{wall} are the thickness and thermal conductivity of the outer wall.

The end caps are treated as annular plates, and the average temperature at the surface facing the gap is defined as:

$$T_{avg} = \frac{T_e + T_c}{2} \quad (10)$$

The node wise energy balance for the end caps is written as:

$$\frac{T_{avg} - T_{s,side}}{R_{th,side}} = h_a A_{side} (T_{s,side} - T_\infty) + \epsilon_{ext} \sigma A_{side} (T_{s,side}^4 - T_\infty^4) \quad (11)$$

The conductive thermal resistance of the end cap insulation layer and the annular area of the end caps are given by:

$$R_{th,side} = \frac{\delta_{ins}}{A_{side} K_{ins}} \quad (12)$$

$$A_{side} = \pi((r_c + \delta_{wall})^2 - r_e^2) \quad (13)$$

To consistently evaluate the areal water productivity and energy utilization of the single-stage unit, the solar irradiation area used for calculation is defined as:

$$A_{nom} = \pi r_e^2 \quad (14)$$

The nominal evaporation rate is further defined as:

$$v_{nom} = \frac{J \bullet 3600}{A_{nom}} \quad (15)$$

The total efficiency, defined as the fraction of the input energy converted into effective latent heat of evaporation, is calculated as:

$$\eta_{tot} = \frac{Q_{evap}}{Q_{in}} \times 100\% \quad (16)$$

The recoverable efficiency is defined as the ratio of the net effective latent heat to the total input energy. By accounting for sidewall heat losses, this metric quantifies the retain latent heat for driving subsequent cascaded evaporation:

$$\eta_{rec} = \frac{Q_{evap} - Q_{loss,side}}{Q_{in}} \times 100\% \quad (17)$$

It should be noted that the total efficiency (η_{tot}) and recoverable efficiency (η_{rec}) defined in this study are heat-utilization indices developed to characterize the evaporation and recoverable heat-transfer processes, respectively. For a single-stage unit, both indices retain the conventional percentage form, and their cumulative extensions in the multi-stage system account for repeated utilization of condensation latent heat.

As shown in Fig. 3b, the nonlinear equation of the single-stage model is solved using a double nested iterative scheme. In the outer loop, the input parameters (Q_{in} , T_∞ , L , r_e , r_c , δ_{gap} , δ_{wall} , k_{ins} , k_{gas} , ϵ_{ext} , h_a) are specified, and the condensation-interface temperature T_c is initialized. In the inner loop, T_c is held fixed and the evaporation-interface temperature T_e is determined by a bisection search to satisfy the local radial energy balance across the gap, $Q_{evap} + Q_{cond} + Q_{rad}$ as evaluated from T_e and T_c . After convergence of the inner loop, the heat losses through the outer wall ($Q_{loss,out}$) and the end caps ($Q_{loss,side}$) are calculated, and a global energy residual is evaluated as $\Delta Q = Q_{in} - (Q_{loss,out} + Q_{loss,side})$. If $|\Delta Q| > \delta$, T_c is updated according to the residual result and the nested iteration is repeated until both the local radial balance and the global energy conservation are satisfied, and the solution yields T_e , T_c , and the corresponding heat fluxes and evaporation performance metrics.

2.3.2. Multi-stage heat and mass transfer model

Based on the single-stage model, a steady-state one-dimensional

radial model is further developed for the multi-stage coaxially nested configuration to quantify cascaded latent heat recovery and cumulative freshwater production. As shown in Fig. 4a, the distiller consists of n similar evaporation–condensation units nested coaxially in the radial direction. For any stage i , vapor diffusion across the vapor gap as well as conductive and radiative heat transfer are described using the previous mechanisms and formulations as in the single-stage model. The model incorporates a critical extension regarding interstage coupling, wherein the condensation heat from stage i acts as the effective input for stage $i + 1$ via substrate conduction. By enforcing temperature and heat flow continuity at the interface, the cascaded energy network is effectively resolved.

Based on the coaxial nesting configuration, the evaporation radius $r_{e,i}$ and condensation radius $r_{c,i}$ for the i -th stage ($i = 1, 2, \dots, n$) are expressed as:

$$r_{e,i} = r_1 + (i - 1)\delta_{gap} \quad (18)$$

$$r_{c,i} = r_{e,i} + \delta_{gap} \quad (19)$$

where r_1 is the first layer radius, δ_{gap} denotes the vapor gap width, and L represents the distiller length. Accordingly, the radial evaporation area and the annular end-cap area are calculated as:

$$A_{e,i} = 2\pi r_{e,i} L \quad (20)$$

$$A_{side,i} = \pi(r_{c,i}^2 - r_{e,i}^2) \quad (21)$$

For a generic stage i , the steady-state energy balance dictates that the input energy at the evaporation surface is dissipated via three pathways, expressed as:

$$Q_{in,i} = Q_{radial,i} = Q_{evap,i} + Q_{cond,i} + Q_{rad,i} \quad (22)$$

where $Q_{evap,i}$, $Q_{cond,i}$, and $Q_{rad,i}$ denote the evaporative latent heat, conductive heat loss, and radiative heat flux of the i -th stage, respectively, which can be directly calculated using Eqs. (2–6) in Section 2.3.1.

According to energy conservation, the radial energy reaching the condensation side of stage i , minus the sidewall loss, yields the effective output energy:

$$Q_{out,i} = Q_{radial,i} - Q_{side,i} \quad (23)$$

where $Q_{side,i}$ represents the sidewall heat loss corresponding to the i -th stage, calculated according to Eqs. (10–13).

Due to the negligible thickness of the evaporation and condensation layers, the contact thermal resistance is disregarded. Consequently, a temperature continuity condition is imposed at the contact interface:

$$T_{c,i} = T_{e,i+1} \quad i = 1, 2, \dots, n - 1 \quad (24)$$

Alongside temperature continuity, heat flux continuity implies that the output energy from the i -th stage equals the input energy to the $(i + 1)$ -th stage:

$$Q_{out,i} = Q_{in,i+1} \quad i = 1, 2, \dots, n - 1 \quad (25)$$

The total energy input is applied exclusively to the first stage, given by the superposition of photothermal and magnetothermal heating:

$$Q_{in,1} = Q_{sun} + Q_{mag} \quad (26)$$

For the final stage acting as the system outlet, the net available energy is balanced by the outer wall heat loss, expressed as:

$$Q_{out,n} = Q_{in,n} - Q_{side,n} \quad (27)$$

where $Q_{out,n}$ is determined by imposing the third-kind boundary condition, resolved via the nodal energy balance Eqs. (7–9).

Similarly, the total evaporation mass flux for the multi-stage system is defined as:

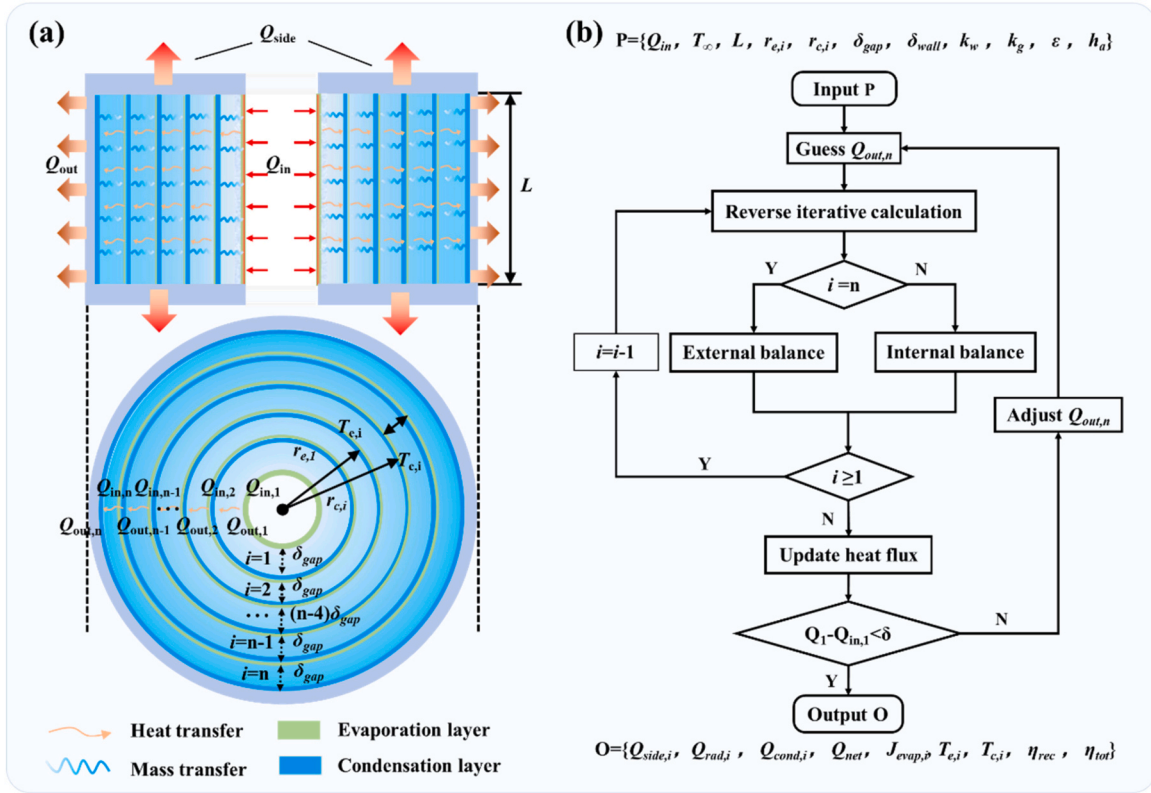


Fig. 4. Heat and mass transfer model for the multi-stage coaxial nested distiller. (a) Schematic illustration of the heat transfer model. (b) Computational flowchart for solving the heat and mass transfer equations.

$$J_{tot} = \sum_{i=1}^n J_{evap,i} \quad (28)$$

Accordingly, the corresponding actual evaporation rate (v_{act}^n) and nominal evaporation rate (v_{nom}^n) are determined by:

$$v_{act}^n = \frac{J_{tot} \cdot 3600}{A} \quad (29)$$

$$v_{nom}^n = \frac{J_{tot} \cdot 3600}{A_{nom}} \quad (30)$$

The cumulative total efficiency is calculated based on the evaporative latent heat of each stage:

$$\eta_{tot}^n = \frac{\sum_{i=1}^n Q_{evap,i}}{Q_{in,1}} \times 100\% \quad (31)$$

Additionally, the cumulative recovery efficiency of the multi-stage system is defined as:

$$\eta_{rec}^n = \frac{\sum_{i=1}^n (Q_{evap,i} - Q_{side,i})}{Q_{in,1}} \times 100\% \quad (32)$$

To facilitate comparison with conventional thermal desalination systems, the gained output ratio (GOR) is additionally introduced. For the multi-stage MSCSMD, GOR is defined as the cumulative latent heat associated with freshwater production divided by the external energy supplied to the first stage:

$$GOR_n = \frac{\sum_{i=1}^n Q_{evap,i}}{Q_{in,1}} \quad (33)$$

The heat transferred from stage i to stage $i+1$ is recovered internal

energy and is therefore not counted as an additional external input. Under the same energy basis, GOR provides a conventional dimensionless representation of the cumulative latent-heat utilization.

Due to the inherent coupling of thermodynamic parameters and the sensitivity of boundary conditions, a forward solving approach is computationally unstable. Consequently, a reverse iterative scheme is developed to resolve this multi-boundary value problem (Fig. 4b). The algorithm begins by initializing the parameter set P and estimating the outermost heat loss $Q_{out,n}$ to decouple the system. A backward recurrence ($i = n \rightarrow 1$) is then performed: for the final stage $T_{c,n}$ is derived from the wall cooling model, for internal stages ($1 \leq i < n$), the condensation boundary is fixed by enforcing temperature continuity ($T_{c,i} = T_{e,i+1}$). The operating temperature $T_{e,i}$ and input flux $Q_{in,i}$ are then solved via local energy balance equations. Upon reaching the first stage, the algorithm checks for global convergence by comparing the calculated input $Q_{in,1}$ with the actual source power Q_1 . If the residual δ is outside the tolerance range, the guess $Q_{out,n}$ is updated, and the back-calculation repeats until the global energy balance is met, outputting the final steady-state dataset O .

2.4. Model validation

To validate the reliability of the theoretical model, experimental verification was conducted using 1–3 stage device configurations under synergistic alternating magnetic field (AMF) and solar heating conditions. Based on the parameters in Table S1, theoretical calculations were conducted, and the theoretical predictions for outer wall temperature and hourly (1 h) water yield were rigorously compared with experimental results. As depicted in Fig. 5a, regarding the thermal distribution, the model accurately reproduces the decreasing trend of the outer wall temperature with increasing stage number. Specifically, the calculated outer wall temperatures for the 1, 2, and 3 stages configurations are 35.84, 32.67, and 30.31 °C, respectively, compared to the

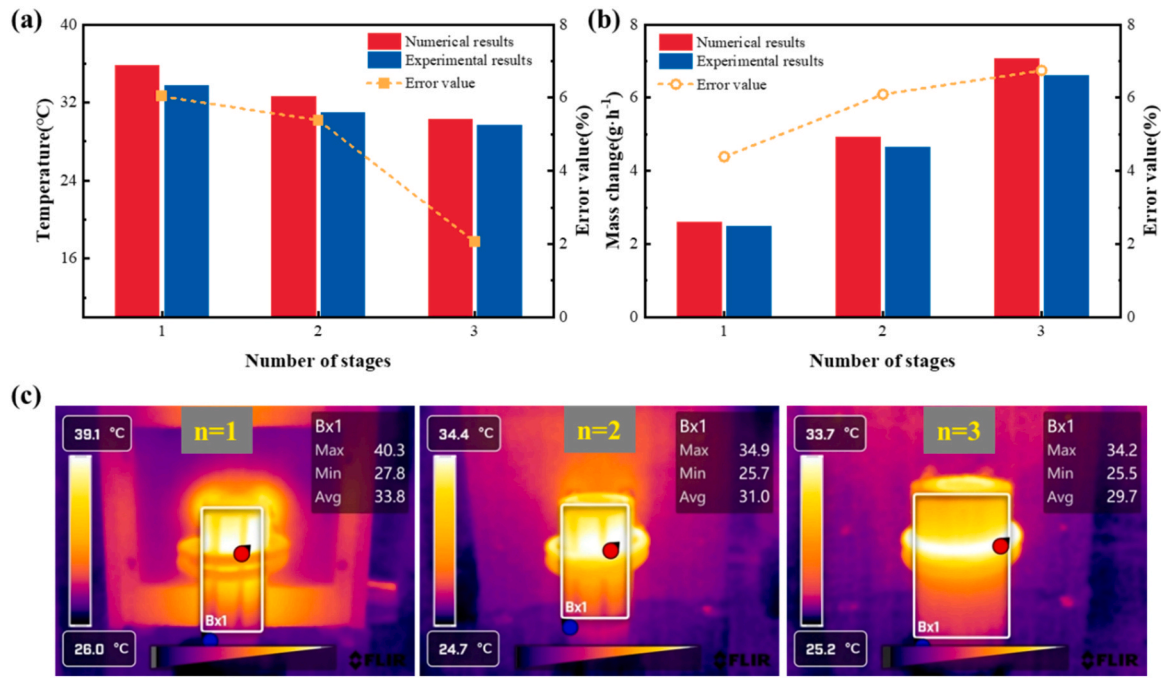


Fig. 5. Model validation and experimental comparison under synergistic AMF and solar heating conditions. (a) Comparison between theoretical predictions and experimental results of the outer wall temperature for 1–3 stage distillers, including relative errors; (b) Comparison of theoretical and experimental results for the 1-h cumulative water yield, with relative errors indicated; (c) Infrared thermal images of the distillers for $n = 1, 2, 3$, illustrating the outer wall temperature distribution and the temperature measurement regions.

experimental values of 33.80, 31.00, and 29.70 °C, the corresponding relative deviations are 6.04%, 5.39%, and 2.05%. Consistently, the infrared thermal images in Fig. 5c visualize a reduction in the average outer wall temperature as the stage number increases, aligning well with the model predictions, this corroborates that the modeling of radial heat transfer and boundary heat dissipation effectively captures the thermal response of the distiller.

In terms of water productivity, as shown in Fig. 5b, the model successfully predicts the stage-wise gain trend observed in experiments and maintains high quantitative accuracy. Based on a solar irradiation area of $A = 3.14 \times 10^{-4} \text{ m}^2$, the predicted nominal water yields for the 1- to 3-stage configurations are 2.60, 4.94, and 7.08 g, respectively. In comparison, the measured values are 2.49, 4.66, and 6.63 g, yielding relative errors of 4.38%, 6.09%, and 6.74%. Overall, the deviations for both outer wall temperature and water yield are controlled within a range of approximately 2–7%. The model demonstrates the capability to simultaneously capture the coupled characteristics of decreasing wall temperature and increasing water yield in multi-stage configurations. This confirms that the proposed heat and mass transfer model possesses robust predictive capability under synergistic solar-AMF conditions, establishing a valid foundation for subsequent parametric sensitivity analysis and structural optimization.

3. Results and discussion

This section primarily analyzes the heat and mass transfer processes involved in both single-stage and multi-stage distillers, the two efficiency indices are retained to distinguish total latent-heat utilization from the heat remaining available for cascade recovery, whereas GOR is additionally employed as a conventional desalination metric for evaluating the overall utilization of the external heat input. On this basis, the study investigates and discusses the influence of geometric dimensions (gap width and length), energy input (solar intensity and alternating magnetic field strength), and external environmental conditions (ambient temperature and heat transfer coefficient) on the evaporation performance and energy efficiency.

3.1. Single-stage distiller analysis

The gap width δ_{gap} is an important geometric factor, directly effecting the vapor diffusion distance and thermal resistance. Fig. 6 shows the performance evolution of the single-stage distiller driven by hybrid solar-magnetic energy, with the δ_{gap} varying from 0.01 cm to 3.0 cm. As shown in Fig. 6a, the actual and nominal evaporation rates display a significant monotonic decline as the δ_{gap} increases, particularly for $\delta_{gap} < 0.5 \text{ cm}$ where sensitivity is high. Consistent with Stefan's diffusion theory, a wider δ_{gap} significantly increases vapor diffusion resistance, thus the dominance of this geometric resistance leads to a drastic reduction in water yield per unit area. Simultaneously, increasing the δ_{gap} significantly enhances the thermal resistance, modifying the temperature distribution (Fig. 6b). While T_{evap} remains stable ($>41^\circ\text{C}$), T_{cond} decreases drastically from 43°C to 23°C . This demonstrates that the wider δ_{gap} effectively blocks outward heat conduction, trapping heat within the evaporation layer to maintain a high evaporation temperature, while the condensation layer cools rapidly toward ambient temperature due to the lack of heat supply. Furthermore, with the increase of δ_{gap} , the outer wall temperature closely approaches T_{cond} , the temperature within the side wall reflects the average of the evaporation and condensation temperatures, serving as an indicator of the side wall heat loss trend. The side wall heat loss Q_{side} shows a dramatic near-linear increase, dominating the energy flow after $\delta_{gap} > 1.5 \text{ cm}$ due to the quadratic expansion of the side cover area A_{side} and the resulting high environmental dissipation. Moreover, the wider δ_{gap} expands the temperature difference between T_{evap} and T_{cond} , causing a counter-intuitive rise in internal radiation Q_{rad} which clarifies the redirection of energy transmission. As shown in Fig. 6d, η_{tot} shows a mild decline with δ_{gap} , remaining at 56.99% at 3 cm. However, η_{rec} drops dramatically from 88.72% to 0.91%. Thus, for multi-stage applications, a gap width not exceeding 5 mm is essential to preserve interstage energy recovery and enable efficient coupling.

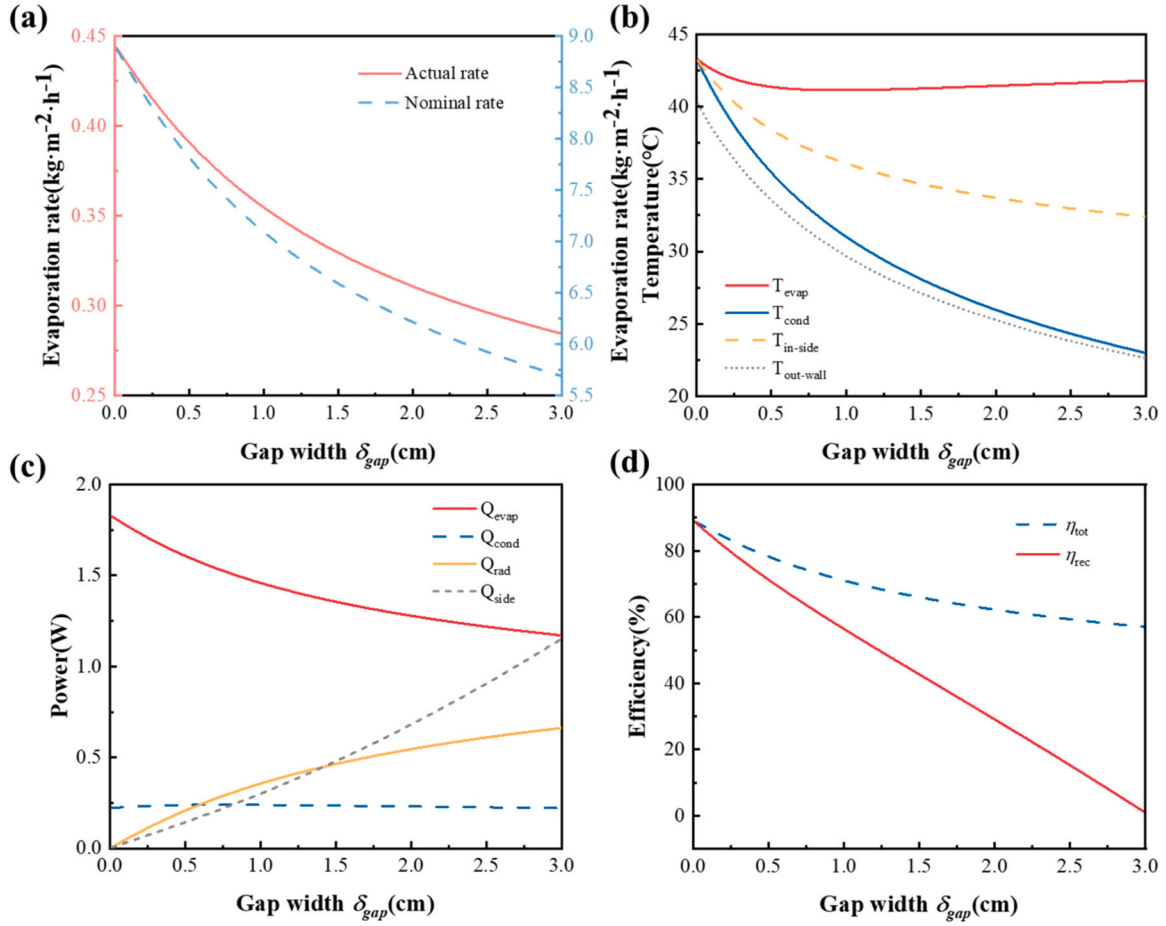


Fig. 6. Impact of δ_{gap} on the performance of the single-stage hybrid distiller. (a) Actual vs. nominal evaporation rates as δ_{gap} increase. (b) Steady state temperature changes for the evaporation, condensation, and side wall. (c) Variation of energy fluxes, including evaporation, conduction, radiation, and sidewall loss. (d) Variation of total efficiency η_{tot} and recovery efficiency η_{rec} to changes in δ_{gap} .

3.2. Multi-stage distiller analysis

Fig. 7 illustrates the performance evolution of the multi-stage distiller operating across 1–16 stages, encompassing the evaporation rate, temperature distribution, and stage-wise heat flux. As shown in Fig. 7a, the evaporation performance displays a marked exponential decay across the stages. Despite consistent trends, there is an order of magnitude difference between nominal and actual rates, where the nominal rate falls from 8.29 to $1.52 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$, while the actual rate declines from $0.41 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ to near zero. This appearance signifies the gradual weakening of the driving force responsible for mass transfer. Furthermore, Fig. 7b elucidates the reasons for this performance decay by showing the distribution of T_{evap} and T_{cond} , while T_{evap} remains above T_{cond} to drive phase change, both curves decline monotonically, and the ΔT narrows with increasing stages. The first stage gives a maximum driving force of 4°C , justifying the highest evaporation rate. However, at the 16th stage, the significantly reduced ΔT leads to a lack of driving force, severely restricting the water evaporation. Fig. 7c quantifies the specific energy distribution, including input power Q_{input} , effective latent heat of evaporation Q_{evap} , as well as heat losses via conduction Q_{cond} , radiation Q_{rad} , and side walls Q_{side} . Results show that Q_{input} declines stage by stage, despite Q_{evap} acting as the dominant term throughout, unavoidable side wall losses reduce the energy available for the next stage. Significantly, the 16th stage shows a distinct heat dissipation Q_{out} , indicating the dissipation of remaining energy. Thus, the MSCSMD performs better in the earlier stages, with the decline in later stages mainly due to the reduced temperature gradient and the weakening of

energy transfer efficiency by heat losses.

Fig. 8 systematically evaluates the influence of the total stage number of the MSCSMD on the temperature distribution, water production rate, energy efficiency, and heat loss pathways. As shown in Fig. 8a, while the first-stage temperatures ($T_{evap,1}$ and $T_{cond,1}$) maintain a stable value around 41°C with only a marginal rise, the outer wall temperature $T_{out-wall}$ undergoes a significant exponential decline, dropping from 35.85°C (1-stage) to 21.00°C (16 stages). This temperature drop demonstrates that the multi-stage configuration effectively expands the overall temperature difference within the system. By enabling the cold wall to approach the ambient temperature, the system fully exploits the potential of the available thermal gradient. As quantified in Fig. 8b, the cumulative evaporation performance versus total stage number. Both nominal and actual rates increase non-linearly, but the growth gradient flattens, indicating a clear marginal diminishing effect. This aligns with the driving force decay shown in Fig. 7, proving that the contribution of additional downstream stages to the total yield decreases as the stage number rises. Nevertheless, the superiority of the MSCSMD is fully substantiated by the efficiency analysis in Fig. 8c. Both the cumulative total efficiency η_{tot} and recoverable efficiency η_{rec} increase with stage number, reflecting the progressive reuse of condensation latent heat within the multi-stage system. For the 16-stage configuration, η_{tot} reaches approximately 650%, corresponding to a GOR of 6.50, while η_{rec} reaches approximately 580%, which serves as compelling proof that the MSCSMD successfully realizes the multiple cascading recovery of latent heat. Fig. 8d depicts the energy pathways for cumulative losses and effective energy. Q_{evap} increases synchronously with Q_{input} as stages are increased, becoming the dominant energy component. Most

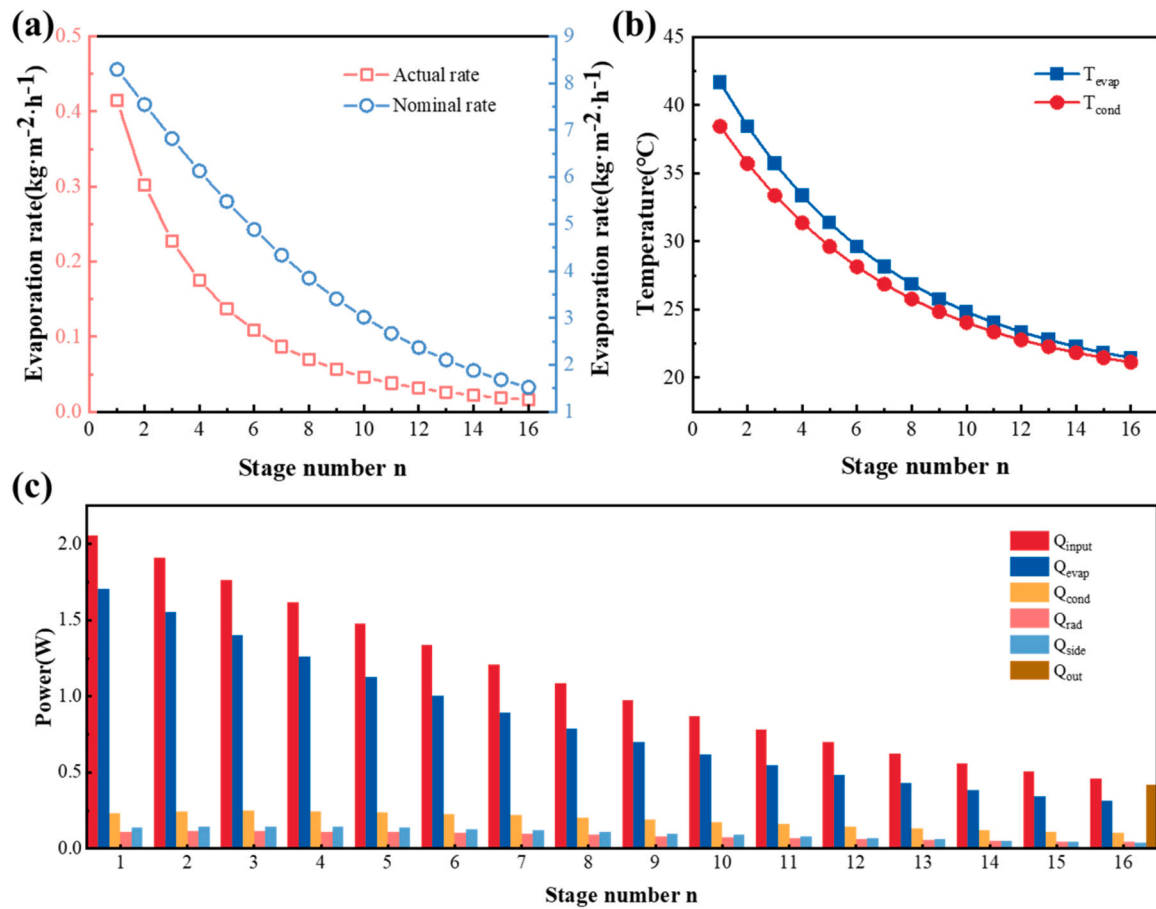


Fig. 7. Detailed thermal performance analysis of the MSCSMD along the stage number (n). (a) Comparison between actual and nominal evaporation rates; (b) Temperature distribution of the evaporation and condensation layers; (c) Breakdown of heat flux components (Q_{input} , Q_{evap} , Q_{cond} , Q_{rad}) and heat losses (Q_{side} , Q_{out}) for each stage.

importantly, the waste heat discharge Q_{out} drops significantly from a high 1-stage value to a minimum at 16 stages, indicating that increasing the stage number reduces terminal heat dissipation. However, the increased surface area leads to a monotonic rise in Q_{side} and Q_{rad} , this suggests an optimal stage limit where the benefits of further latent heat recovery are outweighed by the additional heat dissipation induced by the larger geometry.

3.3. Effect of distiller length

Further research was conducted to examine the impact of the MSCSMD length on evaporation performance and energy efficiency. As shown in Fig. 9a, with L increasing from 4 cm to 20 cm, this section analyzes the regulatory effect of the length on both the thermal performance and the multi-stage evaporation-condensation process. As shown in Fig. 9b, increasing L causes $T_{out-wall}$ to decrease from 54.85 °C at $L = 4$ cm to 28.32 °C at $L = 20$ cm. This is due to the larger heat transfer area dispersing the heat flux, reducing the overall temperature. Additionally, regardless of length, all curves converge near the ambient temperature as n increases, proving that increasing stages effectively neutralizes the terminal heat dissipation caused by distiller geometry. As shown in Fig. 9c, length L has a decisive impact on $T_{e,1}$ and $T_{c,1}$, both of which decrease monotonically with increasing L . At $L = 4$ cm, the high heat flux maintains $T_{e,1}$ at about 65 °C, whereas at $L = 20$ cm, the temperature decreases to approximately 30 °C. Although this reduces the evaporation driving force, the lower operating temperature helps reduce radiative and conductive heat losses. Moreover, Fig. 9d investigates the role of MSCSMD length in boosting efficiency, confirming a remarkable

surge in both η_{tot} and η_{rec} as L expands from 4 cm to 20 cm. For the 16-stage system, η_{tot} increases from 509.36% to 779.77%, corresponding to an increase in GOR from 5.09 to 7.80, while η_{rec} increases from 414.55% to 718.72%. The findings suggest that the enlarged axial length and heat transfer area facilitate a reduction in average operating temperature. Consequently, this mitigates heat dissipation to the ambient environment, especially radiative loss, allowing a greater portion of input energy to be retained within the device for latent heat recycling, thus realizing optimal energy efficiency. However, considering practical limitations such as the magnetic heating coverage and the water transport distance, the MSCSMD cannot be extended indefinitely. Thus, to balance performance and feasibility, the length is fixed at 10 cm in the following research.

As depicted in Fig. 10a, the actual evaporation rate shows a clear dependence on length. The shortest length ($L = 4$ cm) yields the maximum flux of 3.86 kg·m⁻²·h⁻¹, while increasing L to 20 cm reduces the rate to 0.97 kg·m⁻²·h⁻¹. This inverse trend results from the higher temperatures in shorter units (Fig. 9c), which enhance the mass transfer driving force. Interestingly, Fig. 10b shows a contrasting trend for the nominal rate, which rises to 77.86 kg·m⁻²·h⁻¹ at $L = 20$ cm, this is evidently attributed to the relatively small solar irradiation area. Although a longer length boosts heat recovery efficiency, the resulting lower temperature reduces the water yield per unit area. Nonetheless, this finding highlights the spatial efficiency of the MSCSMD design. Fig. 10c provides a quantitative analysis of the energy allocation in the 16-stage system. A marked elevation in the total processed heat flux is observed as L increases. Q_{evap} remains the predominant component in all settings, escalating from 10.46 W to 16.01 W. Further evidence from the

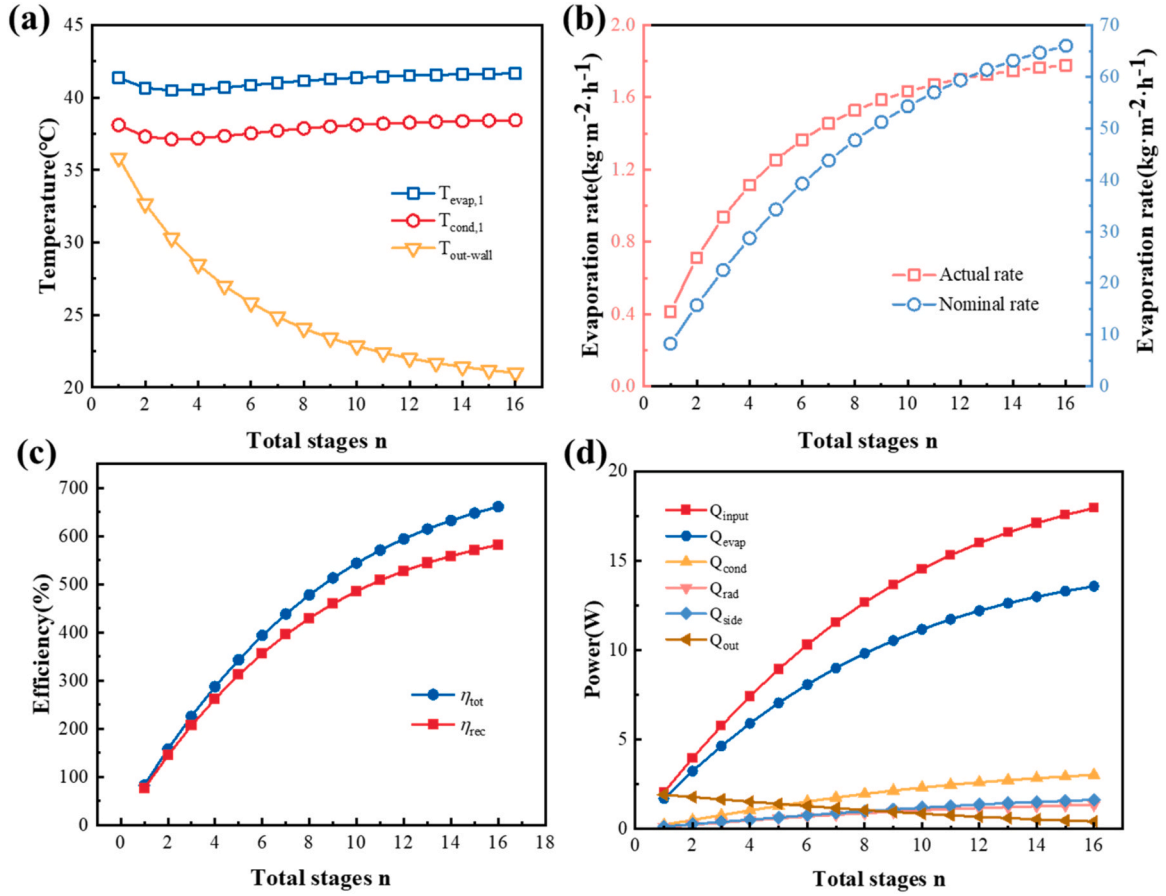


Fig. 8. Evolution of the MSCSMD's cumulative performance across different total stages. (a) Variation of 1st-stage temperatures and outer wall temperature. (b) Growth trends of actual and nominal cumulative evaporation rates. (c) Dependence of total and recovery efficiencies on stage number. (d) Cumulative energy balance and heat loss distribution.

detailed evolution in Figure S1 confirms that while Q_{cond} and Q_{rad} rise in tandem with stage number and length, their relative shares remain limited. Moreover, at $L = 20$ cm, conductive and radiative losses increase to substantial values of 4.13 W and 1.80 W, indicating that preventing the excessive extension of the MSCSMD is crucial to ensuring a high share of latent heat recovery. During operation, the total input energy undergoes multi-stage evaporation and condensation cycles before ultimately being dissipated via the side walls Q_{out} and the outer wall Q_{side} . Fig. 10d reveals the transition of the dominant heat loss mechanism by comparing the evolutionary trends of Q_{out} and Q_{side} . For all length configurations, Q_{side} exhibits a linear or quasi-exponential growth with increasing stage number, whereas Q_{out} displays a monotonic decline due to the reduction in terminal temperature, and the intersection point of these two curves marks the transition of the primary loss mechanism. Notably, the intersection of these curves, marking the shift in the dominant loss mechanism, shifts significantly to the right as length increases (from stage 4 at $L = 4$ cm to stage 13 at $L = 20$ cm). This demonstrates that increasing the cavity length helps minimize side wall losses, forcing more energy to undergo useful evaporation-condensation before being released through the outer wall.

3.4. Effect of operating conditions

The water production performance of the MSCSMD can be flexibly regulated by adjusting the solar radiation intensity. As illustrated in Fig. 11(a), the evaporation rate of the MSCSMD exhibits a dependence on solar irradiance as the intensity increases from 1 sun to 3 suns. With rising irradiance, both actual and nominal evaporation rates improve significantly. Specifically, for the $n = 16$ configuration, the actual

evaporation rate under 3 suns surpasses $2.0 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$, representing an increase of nearly 60% compared to the 1 sun condition. The performance gain is driven by the temperature rise (Fig. 11b-c). As power increases, $T_{e,1}$ rises from 41°C to 46°C , with synchronous increases in wall temperatures. However, the exponential decay of the wall temperature persists, proving that the multi-stage structure establishes stable thermal gradients. Contrasting with the sharp rise in water yield, the energy efficiency shows only marginal changes, with the curves in Fig. 11d showing high self-similarity across all solar intensities. Fig. 11e provides a quantitative assessment of the efficiency under the 16-stage configuration. From 1 sun to 3 suns, η_{tot} and η_{rec} remain approximately 690.37% and 611.81%, respectively, with the former corresponding to a GOR of approximately 6.90. As a result, with the augmentation of input energy, the effective latent heat flux and the heat transfer processes amplify in tandem and proportionally, this proportional scaling mechanism ensures that the dimensionless efficiency remains fundamentally invariant. Furthermore, the heat transfer pathways are analyzed from the microscopic perspective of energy conservation. As depicted in Fig. 11f, evaporative latent heat, heat conduction, heat radiation, as well as outer and side wall heat losses, all exhibit a proportional growth with solar intensity, while their relative proportions remain stable. Additionally, Figure S2 records the stage-wise cumulative process of energy distribution, confirming that the energy can still be effectively converted into evaporative latent heat even under high solar intensity. Figure S3 further elucidates the interplay between side wall and outer wall heat losses. Notably, despite changes in heat dissipation caused by varying solar irradiance, the cross point between these two losses remains anchored at the 8-th to 9-th stage. This suggests that the determination of the optimal stage number for the MSCSMD is fundamentally dictated

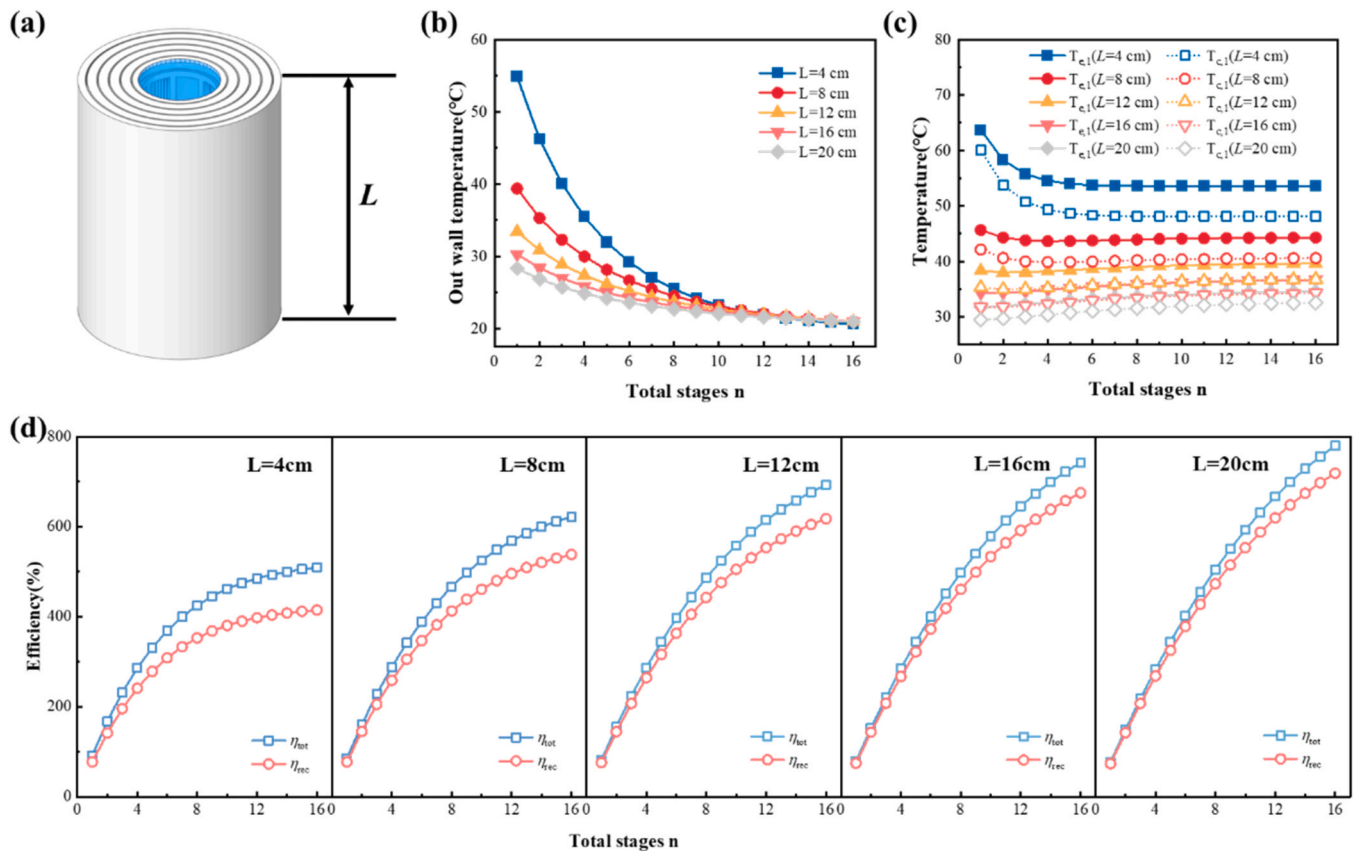


Fig. 9. Thermal performance analysis of the system under varying distiller lengths L . (a) Structural schematic of variable lengths. (b) Outer wall temperature with total stage number for different lengths. (c) First-stage evaporation and condensation temperatures with total stage number for different lengths. (d) Total efficiency and recovery efficiency with total stage number for different lengths $L = 4, 8, 12, 16, 20$ cm.

by its geometric structure, whereas the impact of solar irradiance is negligible.

In addition, the magnetothermal energy can be regulated by adjusting the coil current. For currents of 4.4 A, 6.6 A, and 8.8 A, the resulting input powers are 0.87 W, 1.15 W, and 1.74 W, respectively. As shown in Fig. 12a, as the current increases from 4.4 A to 8.8 A, the magnetothermal effect is significantly enhanced. Consequently, the actual evaporation rate rises from approximately $1.78 \text{ kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$ to $2.41 \text{ kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$, while the corresponding nominal evaporation rate increases from $65.99 \text{ kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$ to $90.02 \text{ kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$. As evidenced by the temperature profiles in Fig. 12b-c, the performance enhancement stems from the enhanced magnetothermal effect caused by higher coil currents. The resulting increased energy input to the magnetic porous carbon forces a sharp increase in the evaporation, condensation, and outer wall temperatures. In contrary to the slightly effect of solar irradiance, Fig. 12d-e show that efficiency increases sharply with current under magnetothermal conditions. Raising the current from 4.4 to 8.8 A increases the cumulative total efficiency from 648.64% to 690.37%, corresponding to an increase in GOR from 6.49 to 6.90. This indicates that increasing the magnetothermal input promotes internal heat transfer and increases the cumulative efficiency under the investigated conditions. In addition, Fig. 12f and Figure S4–5 show that the heat transfers and dissipation trends under magnetothermal input align well with those under solar irradiance, as both represent energy inputs in essence. Thus, adjusting either solar intensity or magnetothermal input allows for the flexible control of freshwater production.

3.5. Effect of environmental conditions

In practical engineering applications, the ambient temperature is

inherently variable. Acting as the heat sink for energy dissipation, it directly dictates the temperature gradient between the outer wall and the surroundings. To this end, the influence of ambient temperature on the evaporation performance and efficiency of the MSCSMD was investigated. As illustrated in Fig. 13a, with the ambient temperature rising from 20°C to 30°C , the evaporation rate of the MSCSMD increases significantly. This observation stands in contrast to the conventional view that a lower cold-end temperature is preferred to establish a steeper temperature gradient. To elucidate this phenomenon, a comprehensive analysis of the temperature distribution is necessary. As indicated in Fig. 13b-c, elevated ambient temperatures raise the outer wall temperature, compelling a simultaneous rise in evaporation and condensation temperatures. While a higher ambient temperature narrows the driving thermal gradient, it concurrently mitigates convective and radiative heat losses, sustaining the MSCSMD within a higher operating temperature regime. Crucially, the mass transfer process is dominated by the gain in saturation vapor pressure associated with higher temperatures. This contribution outweighs the detrimental effect of the reduced terminal condensation gradient, ultimately resulting in a net increase in the evaporation rate. Fig. 13d-e show that raising the ambient temperature from 20°C to 30°C drives the η_{rec} from 581.42% to 704.31%, while η_{tot} shows a corresponding increase. The associated GOR increases from 6.60 to 7.78, accompanied by a comparable rise in evaporation efficiency. This confirms that reducing the external thermal gradient is essentially a heat preservation strategy, which ensures the input energy is utilized primarily for latent heat recovery rather than being dissipated into the environment. As demonstrated in Fig. 13f and Figure S6–7, a rise in ambient temperature boosts the evaporative latent heat, while thermal conduction and radiation conversely diminish. This implies that the energy conserved from reduced environmental heat

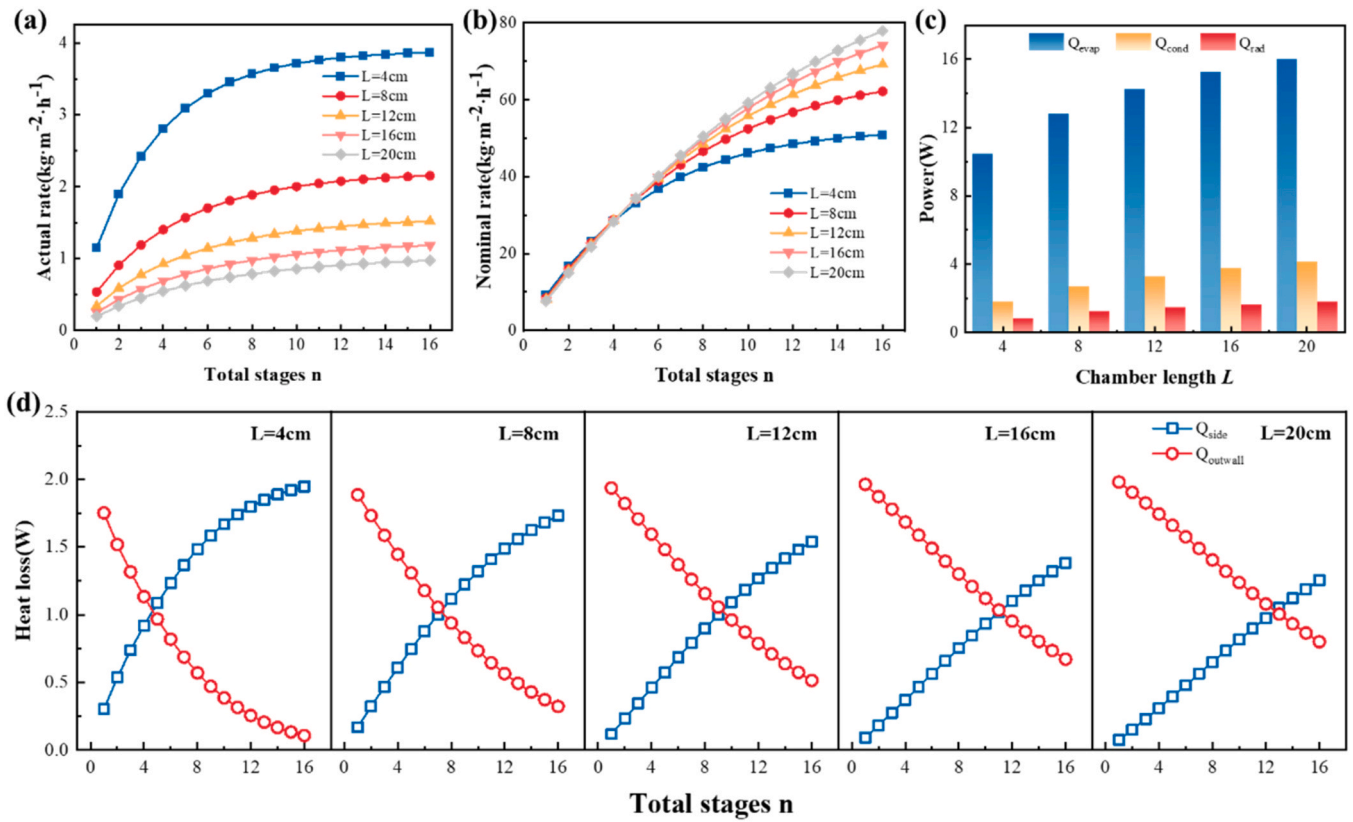


Fig. 10. Influence of distiller length on evaporation rates and heat loss characteristics. (a) Variation of actual cumulative evaporation rate under different lengths; (b) Variation of nominal evaporation rate under different lengths; (c) Comparison of thermal power components (Q_{evap} , Q_{cond} , and Q_{rad}) at the maximum stage number for different lengths; (d) Cumulative side wall heat loss (Q_{side}) and outer wall heat loss (Q_{out}) under different lengths.

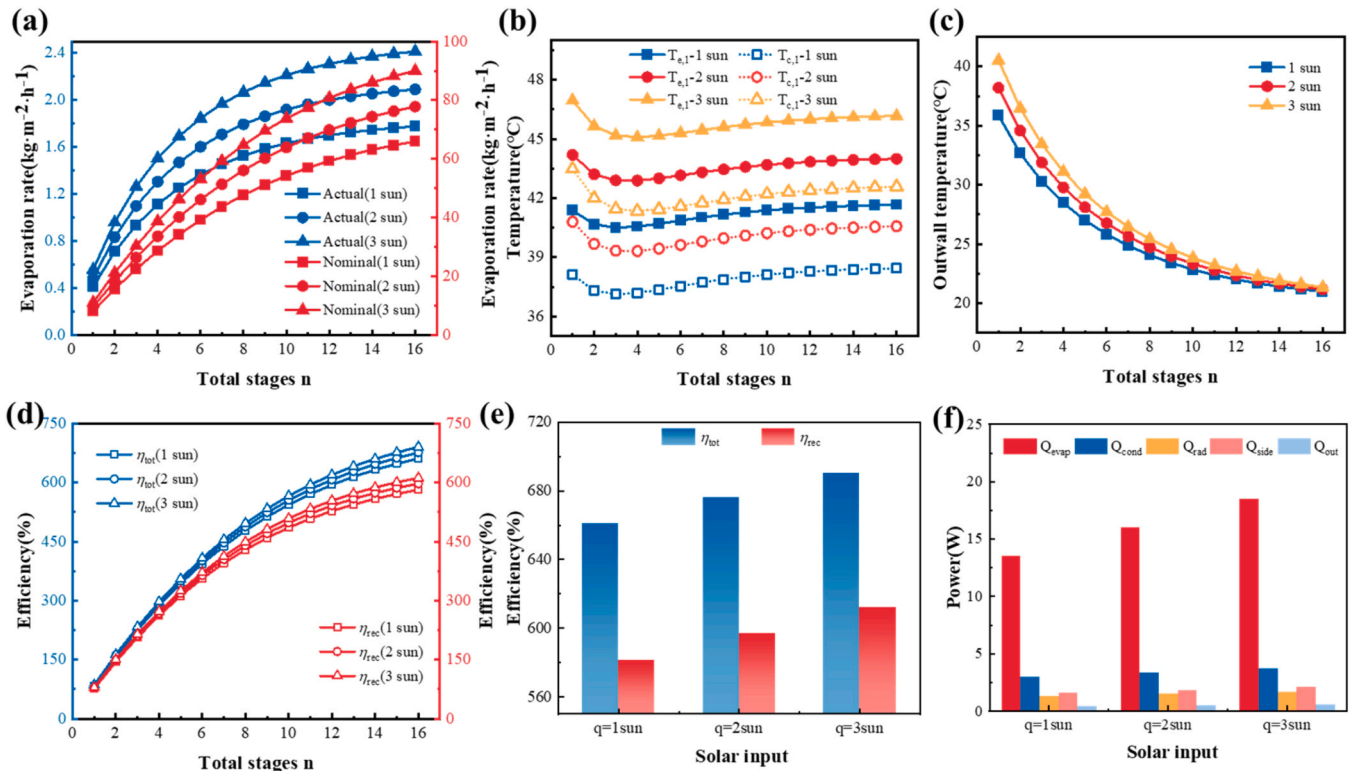


Fig. 11. Influence of solar irradiance on evaporation performance and energy dissipation. (a) Variation of cumulative actual and nominal evaporation rates; (b) Changes in the first-stage evaporation and condensation temperatures; (c) Decay profile of the outer wall temperature; (d) Variation of total efficiency and recovery efficiency; (e) Comparison of efficiencies for the 16-stage configuration; (f) Comparison of heat dissipation components for the 16-stage configuration.

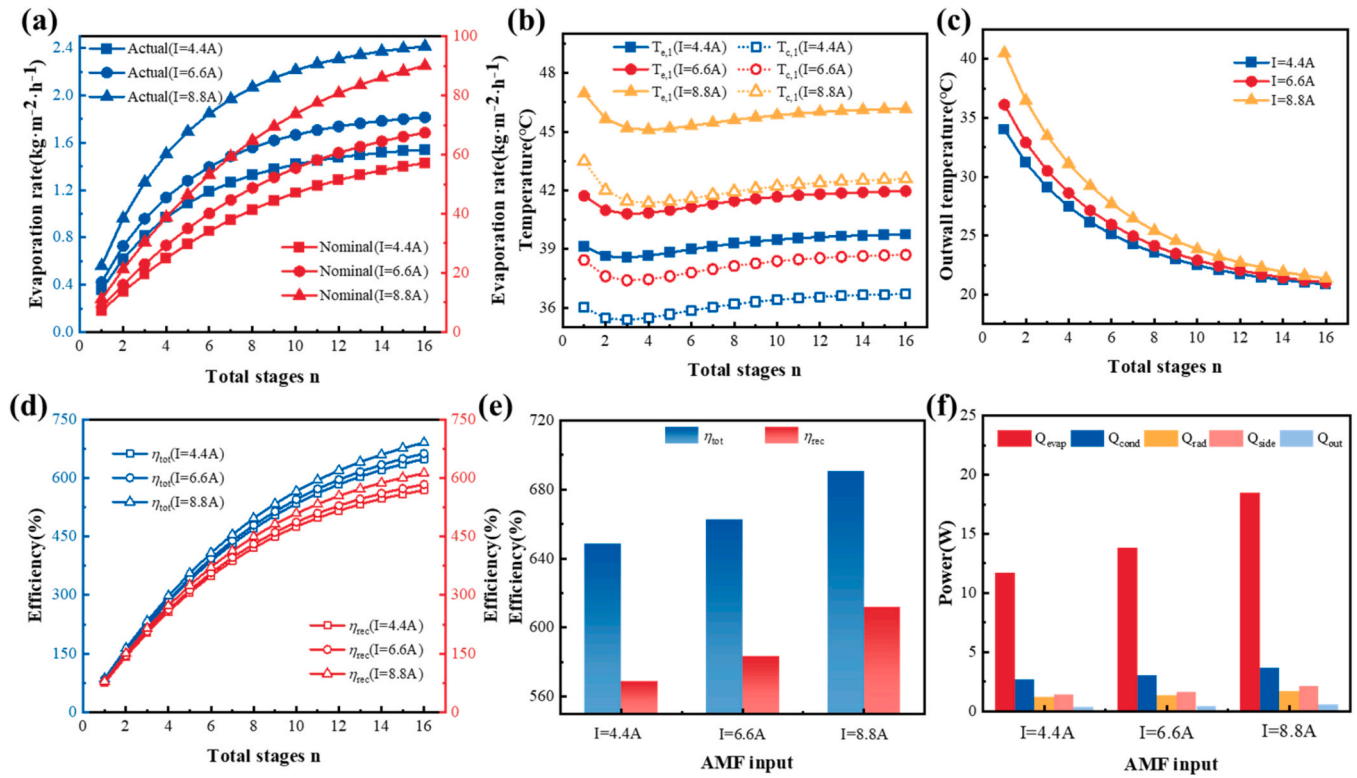


Fig. 12. Performance variations under different alternating magnetic field currents (I=4.4, 6.6, 8.8 A). (a) Variation of cumulative actual and nominal evaporation rates; (b) Variation of first-stage evaporation and condensation temperatures; (c) Variation of outer wall temperature; (d) Variation of total efficiency and recovery efficiency; (e) Efficiency comparison for the 16-stage distiller; (f) Comparison of heat dissipation components for the 16-stage distiller.

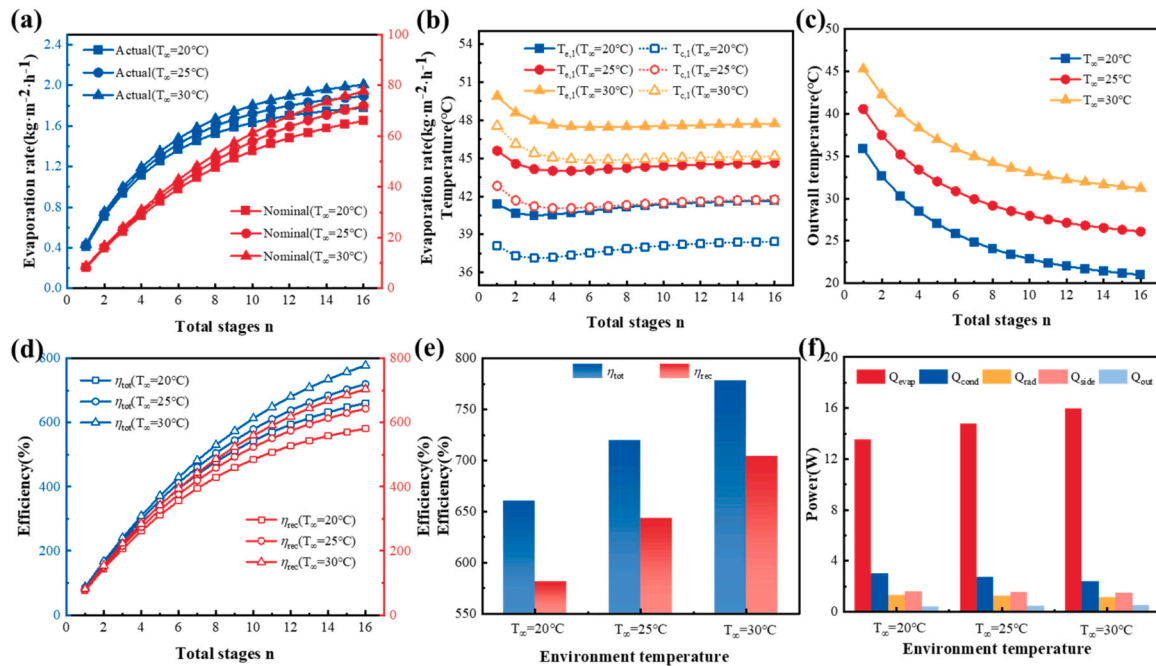


Fig. 13. Influence of ambient temperature (20°C , 25°C , 30°C) on evaporation performance and thermal efficiency. (a) Variation of cumulative actual and nominal evaporation rates with ambient temperature; (b) Variation of first-stage evaporation and condensation temperatures; (c) Variation of outer wall temperature; (d) Variation of total efficiency and recovery efficiency; (e) Comparison of efficiencies for the 16-stage distiller; (f) Comparison of heat dissipation for the 16-stage distiller.

dissipation is not lost through other internal parasitic channels but is effectively channeled into evaporative latent heat. Thus, elevated ambient temperatures reconfigure the heat loss dynamics, realizing a

concurrent enhancement in both distiller efficiency and evaporation rate.

In an open outdoor environment, the distiller is inevitably exposed to

external airflow fluctuations. Since the convective heat transfer coefficient is determined by wind speed and air flow patterns, these environmental factors significantly affect the convective heat dissipation from the side wall and outer wall. As shown in Fig. 14a, a monotonic decrease in evaporation rate is observed with increasing heat transfer coefficient. With h rising from 5 to $15 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$, the actual and nominal rates fall significantly to 1.63 and $58.20 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$, respectively. This proves that strong external convection greatly impairs water productivity by reducing the effective mass transfer flux under the same power input. As shown in Fig. 14b–c, a higher h cools the distiller significantly, dropping the temperature from 45.31°C to 36.63°C . Even though the condensation temperature drops by a similar margin, the pressure difference driving the phase change is severely reduced. Additionally, the rapid decay of the outer wall temperature confirms increased heat loss, meaning heat is swept away by convection before recycled in subsequent stages. A quantitative assessment presented in Fig. 14d–e reveals the dependency of efficiency on convective heat transfer, while the impact is negligible for $n < 4$, the heat loss associated with high heat transfer coefficients amplifies significantly with increasing stage number. In the case of the 16-stage configuration, increasing h from 5 to $15 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ causes the recovery efficiency to drop drastically from 497.85% to 363.32%, accompanied by a commensurate reduction in total efficiency and GOR. Although input energy is constant, Fig. 14f and Figure S8–9 show a marked reduction in effective latent heat, with other loss channels remaining stable. This confirms that the main reason for the efficiency drop is the intensified heat dissipation caused by a higher heat transfer coefficient. This lowers the distiller operating temperature and diminishes the driving force for phase change, thereby restricting the generation of evaporative latent heat.

As shown in Table 1, the proposed MSCSMD shows a higher evaporation rate and total efficiency than the representative solar stills reported in recent studies [26], [29], [31–34], [46]. These systems mainly improve performance by using thermal localization, multistage latent

Table 1

Comparison of water production and efficiency with different solar stills.

Type	Reference	Evaporation rate	Efficiency	GOR
TSM	Gao et al. [26]	$4.74 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	322%	3.22
Localized-TMSS	Zhang et al. [29]	$3.41 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	750%	7.50
ATMSMD	He et al. [31]	$4.32 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	81%	0.81
MHMSS	Zhang et al. [32]	$10.91 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	335.29%	3.35
FISS	Zhao et al. [33]	$6.5 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	480%	4.80
STB-MD	Poredoš et al. [34]	$9.0 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	407%	4.07
TMSS	Xu et al. [46]	$5.78 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	385%	3.85
MSCSMD	This work	$65.99 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$	779.77%	7.80

heat recovery, thermohaline convection, waste heat injection, or mass-transfer optimization. In comparison, the MSCSMD combines a coaxial structure with solar-magnetothermal heating, achieving a nominal evaporation rate of $65.99 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ and a total efficiency of 779.77%, corresponds to a GOR of 7.80 under the defined external-energy basis. Compared with our previous horizontally stacked MHMSS, the coaxial structure reduces the heat loss from the upper surface and forms a more compact radial evaporation–condensation pathway [29]. This helps strengthen latent heat reuse between adjacent stages and improves the overall energy utilization. The high evaporation rate is also related to the nominal-area-based evaluation of the three-dimensional magnetic porous carbon scaffold and the additional heat input from the alternating magnetic field. Overall, the high energy-to-water conversion efficiency indicates the potential of the coaxial solar-magnetothermal design for high-performance freshwater production.

3.6. Economic analysis

To assess the investment potential of the MSCSMD in practical engineering applications, a quantitative analysis of both manufacturing and operating costs was conducted. As detailed in Table S2, the

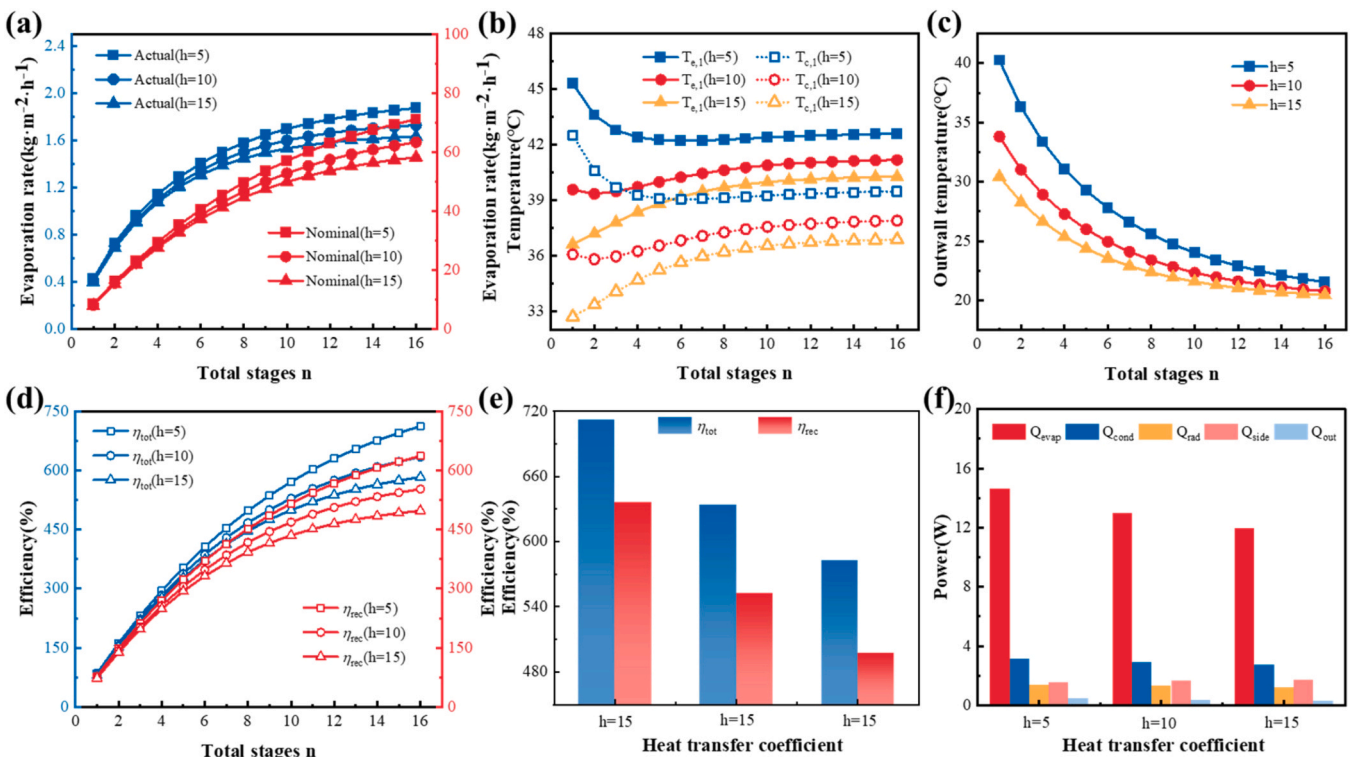


Fig. 14. Effect of heat transfer coefficient ($h=5, 10, 15 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$) on evaporation performance and efficiency. (a) Variation of cumulative actual and nominal evaporation rates; (b) Variation of first-stage evaporation and condensation temperatures; (c) Variation of outer wall temperature; (d) Variation of total efficiency and recovery efficiency; (e) Efficiency comparison for the 16-stage distiller; (f) Comparison of heat distribution for the 16-stage distiller.

manufacturing cost for a single MSCSMD unit stands at 1320 RMB. Notably, the magnetic porous carbon (500 RMB) and the main distiller structure (600 RMB) collectively account for 83% of the total cost. As detailed in Table 2, the MSCSMD demonstrates robust performance in both solar-only and hybrid operation modes. With a daily solar input of 4.5 kWh and a projected efficiency of 600% (equivalent to $GOR=6.0$ under the adopted energy basis), the device delivers a competitive daily yield of $43.09 \text{ L}\cdot\text{m}^{-2}$. Furthermore, by transitioning to the coupled mode with an additional input of 4 kWh of auxiliary electrical energy, a significantly higher daily water yield of $81.38 \text{ L}\cdot\text{m}^{-2}$ is achieved [29]. This signifies that the magnetothermal effect effectively overcomes the production capacity limit inherent to solar-only systems, leading to a significant enhancement in the water production. Despite the extra power cost introduced by the coupled mode, an additional freshwater production of $38.29 \text{ L}\cdot\text{m}^{-2}$ is obtained for an electricity cost of 1.64 RMB. Consequently, the electricity cost for this incremental production stands at only $0.043 \text{ RMB}\cdot\text{L}^{-1}$, which is much below the price levels of commercial bottled water and competitive with industrial desalinated water [47]. In summary, the MSCSMD presents a highly flexible economic strategy that allows for seamless switching between zero-cost and low-cost operational modes in response to real-time water demand or electricity price fluctuations. This capability enables the maximization of freshwater yield while strictly maintaining economic optimality.

4. Conclusions

This work establishes a multi-stage coaxial solar-magnetothermal hybrid distiller (MSCSMD), supported by a validated steady-state model based on global energy balance and a reverse iterative algorithm. Systematic analysis of geometric and operational parameters leads to the following conclusions:

- (1) By leveraging a multi-stage coaxial architecture, the MSCSMD achieves efficient, stepwise recycling of latent heat. For a 16-stage structure, the system attains a recovery efficiency of 581.42% and a cumulative nominal evaporation rate of $65.99 \text{ kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$, equivalent to a GOR of 6.60. Crucially, as the number of stages increases, side-wall heat dissipation gradually surpasses outer-wall heat loss and becomes the dominant heat-loss pathway.
- (2) Gap width critically governs system performance. Although larger gaps increase the temperature difference, the accompanying surge in diffusion resistance and side-wall heat loss causes an exponential decline in recovery efficiency. At a 3 cm gap width, the evaporation rate and recovery efficiency collapse to $0.28 \text{ kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$ and 0.91%, respectively. A gap width of 2.5 mm is identified as optimal for effective heat retention. 65112125
- (3) Device length is a decisive factor in efficiency enhancement. Increasing the length from 4 to 20 cm markedly suppresses side-wall heat dissipation, boosting the total efficiency of the 16-stage system to 779.77%, corresponding to a GOR of 7.80. These results confirm that a large aspect ratio effectively locks internal energy, although practical implementation must account for magnetic field coverage constraints.
- (4) Elevated ambient temperatures and reduced external heat transfer unequivocally improve performance by minimizing environmental heat loss and stabilizing operation in a higher thermal regime. Furthermore, the hybrid operating mode leverages auxiliary power to achieve an exceptionally low freshwater cost of $0.043 \text{ RMB}\cdot\text{L}^{-1}$, underscoring the MSCSMD as a cost-effective and scalable solution for sustainable water production.

CRedit authorship contribution statement

Chenghua Zhang: Writing – original draft, Software, Investigation, Data curation. **Yunfei Yan:** Conceptualization. **Yonghong Wu:** Formal

Table 2

Economic evaluation of magnetic heat-assisted multi-stage solar stills.

Condition	Daily irradiation (kWh)	Daily consumption (kWh)	Energy cost (RMB)	water yield (L)
SUN-only	4.5	/	/	43.09
SUN-AMF	4.5	4	1.64	81.38

analysis, Data curation. **Changlei Qin:** Writing – review & editing, Formal analysis. **Wen Siang Lew:** Writing – review & editing, Formal analysis.

Declaration of Competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Supporting Information

Supporting materials are available depending on the author or publisher.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijheatmasstransfer.2026.129710.

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